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REPORT ON
“SOCIAL MEDIA AND ANALYTICS”

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SUBMITTED BY
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CERTIFICATE

This is to certify that the Audit course

Report Entitled

“SOCIAL MEDIA AND ANALYTICS”

SUBMITTED BY

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Is a bonafide work carried out by students under the supervision of **Prof. Auti M. A.** and it is submitted towards the partial fulfillment of the requirement of third year of Engineering in Artificial intelligence and data science under the Savitribai Phule Pune University during the academic year 2025-2026.

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Your's faithfully

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ABSTRACT

Social media analytics is the process of collecting, analyzing, and interpreting data generated from social media platforms to obtain meaningful insights. This report focuses on studying social media data, user behavior, engagement metrics, and the role of analytics in business decision-making. Social media platforms generate large volumes of structured and unstructured data through posts, comments, likes, shares, and user interactions, which can be analyzed using various tools and techniques.

This report covers important concepts such as social media metrics, data collection methods, analytics tools, sentiment analysis, network analysis, and data visualization. It also explains how social media analytics helps organizations understand customer preferences, identify trends, improve marketing strategies, and support better business decisions. A case study and ethical issues related to privacy and data usage are also included. The report concludes that social media analytics plays a significant role in improving business performance and has strong future scope with emerging technologies such as artificial intelligence and machine learning.

Keywords: Social Media Analytics, Data Analysis, User Engagement, Social Media Metrics, Sentiment Analysis, Data Visualization

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INTRODUCTION

Social media analytics is the process of collecting, measuring, analyzing, and interpreting data from social media platforms to gain useful insights. In today's digital world, platforms like Instagram, Facebook, and LinkedIn generate a large amount of data through posts, comments, likes, shares, and user interactions. This data can be analyzed to understand customer behavior, market trends, and public opinion.

Social media analytics helps organizations measure the performance of their social media activities using different metrics such as engagement rate, reach, impressions, and click-through rate. These metrics help businesses know how users interact with their content and whether their strategies are effective. By analyzing this information, companies can improve their marketing campaigns, increase customer engagement, and make better business decisions. Another important use of social media analytics is trend identification. It helps identify popular topics, hashtags, and customer interests. Businesses can use this information to create relevant content and improve brand visibility. It also supports sentiment analysis, where customer opinions are classified as positive, negative, or neutral to understand public response toward products or services.

Social media analytics is widely used in business, education, healthcare, politics, and e-commerce. It supports data-driven decision-making and helps organizations stay competitive in the digital era. With the use of advanced tools and technologies, social media analytics has become an important part of modern business strategy and has significant future scope in the field of artificial intelligence and data science. It is also useful for competitor analysis. Organizations can compare their social media performance with competitors, identify strengths and weaknesses, and improve their digital presence. This provides a competitive advantage in the market. Another important area is customer feedback analysis. Social media platforms contain reviews, opinions, and discussions that help businesses understand customer satisfaction and improve products or services. This supports better customer relationship management. Social media analytics is also used in campaign performance measurement. It helps track whether marketing campaigns are successful by analyzing reach, engagement, and conversion results. Based on this analysis, companies can optimize future campaigns.

With the use of technologies such as machine learning and artificial intelligence, social media analytics has become more advanced. These technologies help in predicting trends, detecting patterns, and generating automated insights from large datasets. This increases the accuracy and efficiency of analysis.

1.2 Social Media Platforms

1. Instagram :

- Instagram is a popular social media platform used for sharing photos, videos, reels, and stories.
- It generates data through likes, comments, shares, saves, and follower interactions.
- Businesses use Instagram analytics to measure engagement, reach, impressions, and audience behavior.
- It helps identify trending content and improve marketing strategies.

2. Facebook :

- Facebook is a widely used platform for communication, content sharing, and business promotion.
- It generates data through posts, reactions, comments, shares, and page insights.
- Businesses use Facebook analytics to monitor audience engagement and campaign performance.
- It helps in understanding customer interests and improving digital marketing.

3. LinkedIn:

- LinkedIn is a professional networking platform.
- Used for career growth and business communication.
- It generates data through connections, post engagement, profile views, and professional interactions.
- LinkedIn analytics helps analyze audience engagement and content performance.
- It is useful for professional branding and business networking.

TYPES OF DATA IN SOCIAL MEDIA

2.1 Structured Data

Structured data refers to data that is organized in a predefined format and can be easily stored in rows and columns. This type of data is easy to collect, manage, and analyze using databases and analytical tools. In social media, structured data mainly includes numerical and measurable information generated through user activities. Examples of structured data include the number of likes, shares, comments, followers, engagement rate, reach, impressions, and click-through rate. This data helps organizations measure performance, track audience engagement, and evaluate the success of social media campaigns. Since structured data is organized, it is commonly used in reporting, dashboards, and statistical analysis.

2.2 Unstructured Data

Unstructured data refers to data that does not have a fixed format and is more complex to analyze. A large amount of social media data falls into this category. It includes text posts, comments, reviews, images, videos, and multimedia content shared by users. This type of data is useful for understanding customer opinions, public sentiment, and user interests. For example, customer comments can be analyzed to identify whether opinions about a product are positive, negative, or neutral. Images and videos can also provide valuable insights into trends and user behavior. Unstructured data is important in sentiment analysis, content analysis, and trend identification.

2.3 Semi-Structured Data

Semi-structured data contains some organizational elements such as tags, labels, or metadata, but it does not follow a strict relational structure. This type of data combines features of both structured and unstructured data. In social media, examples include hashtags, API responses, JSON data, XML data, user profiles, and metadata related to posts. Hashtags help classify content and identify trends, while metadata can include information such as time of posting, location, and device used. Semi-structured data is useful for filtering, categorization, and advanced data analysis. It supports efficient data processing and integration with analytics systems.

2.4 User-Generated Data

User-generated data is data created directly by users while interacting on social media platforms. It includes posts, comments, reviews, uploaded images, and shared videos. This type of data is highly valuable because it reflects user opinions, preferences, experiences, and feedback. Organizations use user-generated data to understand customer needs, improve products and services, and strengthen customer relationships. It also helps in studying user behavior and identifying market trends. Because it

is created by users, this data provides authentic and real-time insights.

2.5 Interaction Data

Interaction data is generated when users engage with content or interact with other users on social media platforms. It includes likes, shares, comments, mentions, clicks, reposts, and reactions. This data helps measure audience engagement and understand how users respond to content. High interaction often indicates effective content and strong audience interest. Organizations use interaction data to evaluate campaign performance, optimize content strategy, and improve audience targeting. It is an important part of social media performance analysis.

ETHICAL ISSUES IN SOCIAL MEDIA

1. Data Privacy:

- Users share personal information on social media platforms.
- Improper use of this data can violate privacy.
- Organizations should protect user information.
- Data should be collected and stored securely.

2. User Consent:

- User permission should be taken before collecting data.
- Users should know how their data is used.
- Data collection should follow privacy policies.
- Proper consent supports ethical practices.

3. Data Misuse:

- Social media data should not be used for unauthorized purposes.
- Misuse can lead to manipulation or unfair targeting.
- Data should only be used for approved purposes.
- Ethical use prevents misuse of information.

4. Fake Information and Misinformation:

- False information can mislead users.
- Misinformation can create social problems.
- Content should be verified before sharing.
- Monitoring helps reduce fake information.

5. Fake Accounts and Bots:

- Fake accounts can create false engagement.
- Bots can spread misleading content.
- They affect trust and analytics accuracy.
- Detection methods should be used to control them.

6. Security Issues:

- Social media data may face hacking risks.
- Unauthorized access can cause data breaches.
- Security measures should protect user data.
- Confidentiality should be maintained.

7. Confidentiality:

- User information should remain confidential.

- Sensitive data should not be shared without permission.
- Confidentiality improves trust.
- It supports ethical data management.

8. Transparency:

- Data practices should be transparent.
- Users should know how data is used.
- Privacy policies should be clearly explained.
- Transparency improves ethical practices.

9. Bias in Data Analysis:

- Biased data can affect results.
- It may lead to unfair decisions.
- Bias should be reduced in analysis.
- Ethical analytics should support fairness.

10. Cyberbullying and Harassment:

- Social media may be misused for harassment.
- Harmful content can affect users negatively.
- Platforms should control abusive behavior.
- Ethical use includes preventing cyberbullying.

11. Intellectual Property Issues:

- Content should not be copied without permission.
- Copyright rules should be followed.
- Original creators should be credited.
- Ethical use protects intellectual property.

12. Data Accuracy:

- Incorrect data can give wrong conclusions.
- Data should be verified before analysis.
- Accurate data improves reliable insights.
- Ethical analysis depends on data quality.

13. Responsible Use of Analytics:

- Analytics should be used for valid purposes.
- It should not unfairly manipulate users.
- Data should support ethical decisions.
- Responsible use improves trust and fairness.

DATA COLLECTION & PREPROCESSING

1. Data Collection:

Data collection is the process of gathering data from different social media platforms for analysis. It is the first and important step in social media analytics. Data is collected from sources such as posts, comments, likes, shares, hashtags, user profiles, and interactions. The collected data helps organizations understand user behavior, analyze trends, and support decision-making. Effective data collection is necessary to obtain accurate and useful insights from social media platforms.

2 .Data Collection Methods:

Data can be collected using various methods depending on the purpose of analysis. One common method is using APIs provided by social media platforms, which allow access to structured data. Web scraping is another method used to gather publicly available data from websites. Surveys and questionnaires can also be used to collect user opinions and feedback. In addition, social media platforms provide dashboards and built-in analytics tools that offer valuable performance data. These methods help gather relevant information for analysis.

3. Sources of Data:

Social media data can come from different sources such as user posts, comments, likes, shares, hashtags, images, videos, and follower information. Interaction data such as mentions, clicks, and reactions also acts as an important data source. These sources provide both structured and unstructured data, which can be used for performance analysis, sentiment analysis, and trend identification.

Data Preprocessing

1. Data Cleaning:

Data cleaning is the process of improving data quality by removing errors and unwanted information. Duplicate data should be removed, missing values should be handled, and incorrect or noisy data should be corrected. Spam and irrelevant content should also be filtered. Data cleaning improves the accuracy and reliability of analysis.

2. Data Transformation:

Data transformation converts raw data into a suitable format for analysis. This may include normalization, standardization, and feature extraction. Raw social media data often needs transformation before it can be processed by analytical models. This step improves efficiency and supports better analysis.

3. Data Filtering:

Data filtering is the process of selecting useful data and removing irrelevant information. Fake accounts,

spam data, and unnecessary content can affect results, so filtering is important. It helps improve analysis accuracy and ensures only relevant data is used.

4. Data Integration:

Data integration combines data collected from multiple sources into a unified dataset. Integrating data from different platforms provides a more complete analysis and supports better insights. It helps organizations make more informed decisions.

Importance of Data Preprocessing

- ❖ Improves data quality.
- ❖ Removes duplicate and incorrect data.
- ❖ Handles missing values.
- ❖ Filters irrelevant and noisy data.
- ❖ Increases accuracy of analysis.
- ❖ Reduces errors in results.
- ❖ Makes data suitable for processing.
- ❖ Improves efficiency of analytical models.
- ❖ Supports reliable insights.
- ❖ Helps in better decision-making.

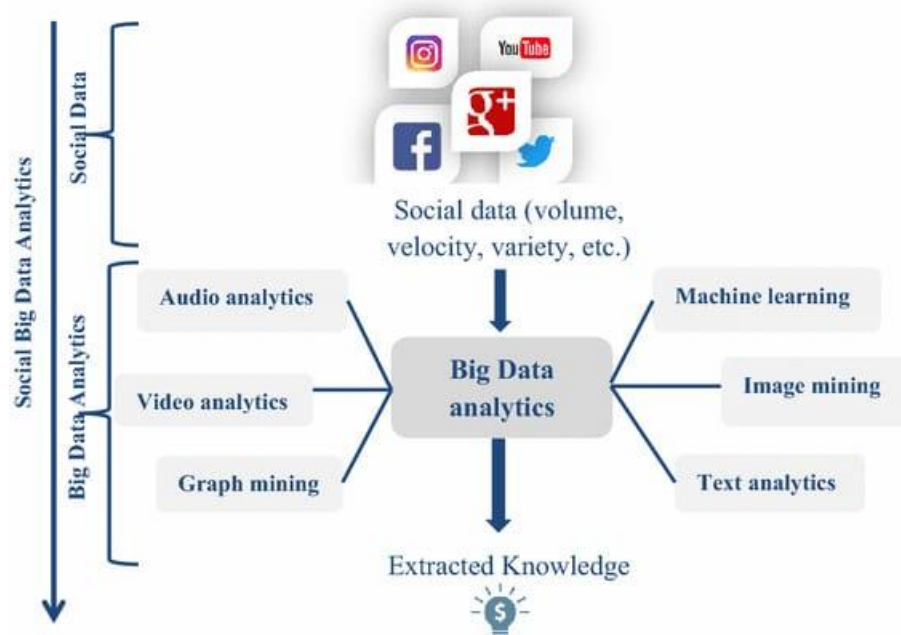


Fig No. 1

MODELING IN SOCIAL MEDIA ANALYTICS

1. Meaning of Modeling:

Modeling in social media analytics refers to the process of applying analytical, statistical, and machine learning models to social media data in order to identify patterns, trends, and relationships. Social media platforms generate large volumes of data through posts, comments, likes, shares, and user interactions. This raw data is complex and difficult to interpret directly, so modeling is used to transform the data into meaningful insights. Modeling helps organizations analyze user behavior, predict trends, classify information, and support decision-making. It plays an important role in improving the effectiveness of social media analysis and business strategies.

2. Classification Models:

Classification models are used to categorize social media data into different classes or groups. These models are widely used in sentiment analysis, where customer opinions are classified as positive, negative, or neutral. Classification models can also be used to identify spam messages, detect fake accounts, classify user interests, and analyze audience behavior. These models help organizations understand different types of data and improve decision-making by providing organized and interpretable results. Classification is an important technique in social media analytics because it supports content analysis and user behavior analysis.

3. Regression Models:

Regression models are used to predict numerical values based on available data. In social media analytics, regression models help predict engagement rates, reach, impressions, follower growth, and campaign performance. For example, a regression model can estimate future engagement based on past social media performance. These models support forecasting, trend analysis, and strategic planning. Regression models help organizations make predictions, optimize content strategies, and improve social media performance through data-driven decisions.

4. Clustering Models:

Clustering models are used to group similar users or data points into clusters based on shared characteristics. In social media analytics, clustering is useful for audience segmentation, identifying groups of users with similar interests, and improving targeted marketing strategies. For example, users can be grouped based on engagement patterns, preferences, or online behavior. Clustering helps organizations better understand their audience, improve personalization, and enhance marketing effectiveness. It is an important modeling technique for customer segmentation and pattern discovery.

5. Predictive Modeling:

Predictive modeling uses historical social media data and analytical techniques to forecast future trends, user behavior, and campaign performance. It helps organizations predict customer responses,

identify upcoming trends, and optimize future strategies. Predictive modeling is useful for planning social media campaigns, improving engagement strategies, and supporting business growth. With the use of artificial intelligence and machine learning, predictive modeling has become more advanced and accurate in social media analytics.

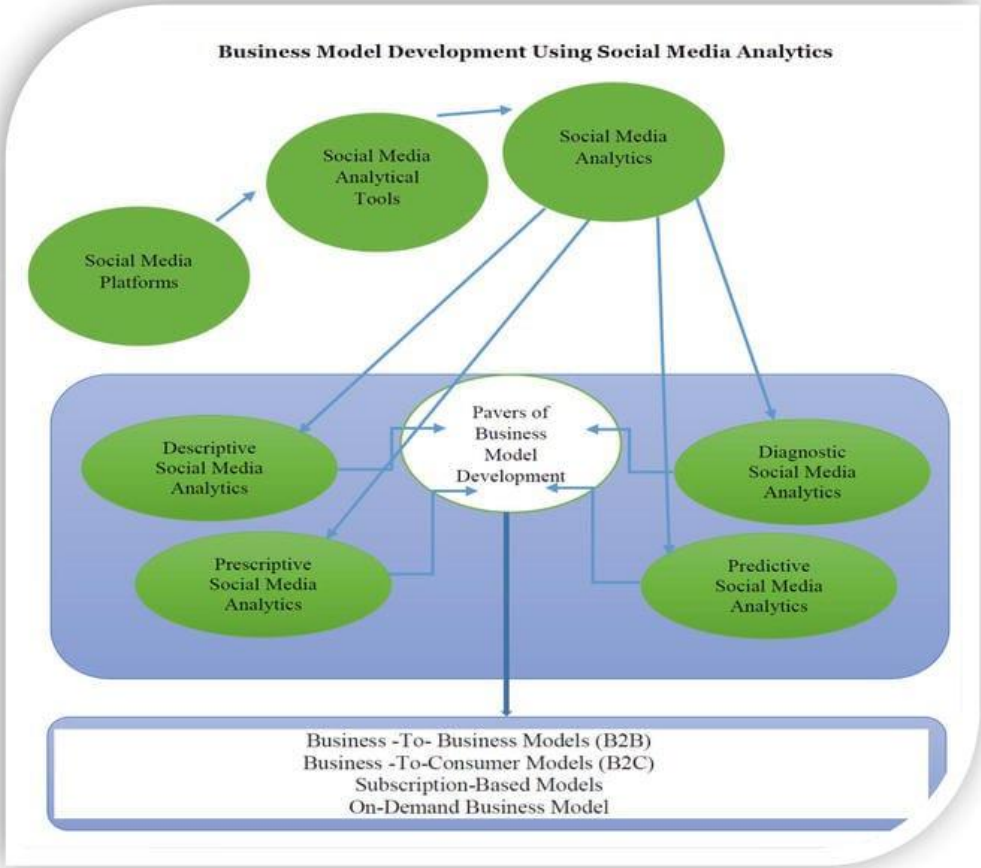


Fig No. 2

SUMMARY STATISTICS AND DATA ANALYSIS

5.1 Summary Statistics:

Summary statistics refers to the process of summarizing and describing social media data using numerical measures to understand its main characteristics. Social media platforms generate large amounts of data such as likes, comments, shares, impressions, and engagement values. Analyzing this raw data directly can be difficult, so summary statistics helps organize and simplify the data for interpretation. It provides a quick understanding of patterns, trends, and variations within the dataset. Common summary statistics include mean, median, mode, minimum, maximum, and standard deviation. These measures are useful for analyzing performance metrics, comparing results, and supporting further data analysis.

5.2 Mean:

Mean represents the average value of a dataset and is calculated by dividing the total of all values by the number of observations. In social media analytics, mean can be used to calculate average likes per post, average comments, average shares, or average engagement rate. It helps provide a general view of social media performance and supports comparison across different posts, campaigns, or platforms. Mean is useful in identifying overall trends and evaluating performance levels.

5.3 Median:

Median represents the middle value in a dataset when all values are arranged in order. It is used to identify the central tendency of data. In social media analytics, median can help analyze typical engagement values without being affected by extremely high or low values. It is particularly useful when data contains outliers, as it provides a more balanced measure compared to the mean. Median helps improve understanding of normal performance patterns.

5.4 Mode:

Mode represents the value that appears most frequently in a dataset. In social media analytics, mode can be used to identify the most common engagement level, most frequent posting time, or most commonly used hashtag. It helps recognize repeated patterns in data and understand common user behavior. Mode is useful when analyzing frequency-based information in social media data.

5.5 Standard Deviation:

Standard deviation measures the variation or spreads of data values from the mean. It helps determine whether data values are close to the average or widely spread. In social media analytics, standar

deviation can be used to analyze consistency in engagement, reach, or performance metrics. A low standard deviation indicates more stable and consistent data, while a high standard deviation shows greater variation. It helps organizations understand performance stability and data distribution.

Data Analysis

Data analysis is the process of examining, interpreting, and analyzing collected social media data to generate useful insights. It involves studying user interactions, engagement metrics, trends, and audience behavior. Data analysis helps organizations identify patterns, measure performance, understand customer preferences, and improve decision-making. Different types of analysis used in social media include performance analysis, sentiment analysis, trend analysis, and audience analysis. It transforms raw data into meaningful information for business strategies.

Importance of Data Analysis:

- Helps in understanding audience demographics and preferences.
- Identifies high-performing and low-performing content.
- Supports better targeting of customers.
- Improves marketing campaign effectiveness.
- Helps in competitor analysis.
- Detects trends and viral content early.
- Supports real-time decision-making.
- Helps in measuring return on investment (ROI).
- Improves customer satisfaction through feedback analysis.
- Helps in risk identification and management.
- Supports data-driven business strategies.

VISUALIZATION OF SOCIAL MEDIA NETWORKS

Visualization:

Visualization is the process of representing data in graphical or visual formats such as charts, graphs, and diagrams to make it easier to understand and analyze. In social media analytics, visualization plays an important role in converting large and complex datasets into simple and meaningful visual representations. Social media platforms generate huge amounts of data through user interactions such as likes, comments, shares, and posts, which can be difficult to interpret in raw form. Visualization helps present this data in a clear and organized manner.

Through visualization, patterns, trends, and relationships in social media data can be easily identified. It allows analysts and organizations to quickly understand user behavior, engagement levels, and performance metrics. Visual tools such as bar charts, line graphs, pie charts, and network diagrams are commonly used to represent different types of data.

Visualization also supports better decision-making by providing insights in a simple and interactive way. It helps businesses monitor their social media performance, analyze marketing campaigns, and improve strategies. Overall, visualization is an essential component of social media analytics as it enhances understanding, improves communication of results, and supports data-driven decisions.

Types of Visualization in Social Media :

1. Engagement Bar Charts

- Used to compare likes, comments, shares, and reactions across posts.
- Helps identify which content performs better.
- Useful for comparing multiple posts or campaigns.

2. Follower Growth Line Graphs

- Show increase or decrease in followers over time.
- Helps track account growth and performance.
- Useful for analyzing long-term trends.

3. Audience Distribution Pie Charts

- Represent audience demographics such as age, gender, and location.
- Help understand target audience.
- Useful for marketing and content planning.

4. Network Graphs

- Show connections between users in a social network.
- Nodes represent users, edges represent interactions.
- Help identify influencers and communities.
- Useful for studying information flow.

5. Hashtag Analysis Charts

- Show popularity and usage of hashtags.
- Help identify trending topics.
- Useful for content optimization.

6. Sentiment Visualization

- Represent user opinions as positive, negative, or neutral.
- Often shown using pie charts or bar graphs.
- Helps understand public opinion about a brand or product.

7. Dashboard Visualization

- Combine multiple visualizations in one place.
- Provide overall view of performance metrics.
- Used for real-time monitoring and decision-making.

8. Video Performance Graphs

- Show views, watch time, and engagement on video content.
- Useful for analyzing video performance on platforms like YouTube.

9. Story/Content Performance Charts

- Analyze performance of stories and short content.
- Commonly used on platforms like Instagram.
- Help improve content strategy.

TOOLS FOR SOCIAL MEDIA VISUALIZATION

1. Tableau

- Used for creating interactive dashboards and reports.
- Supports large datasets and real-time analysis.
- Helps visualize trends, patterns, and performance metrics.
- Easy to use with drag-and-drop features.

2. Power BI

- Used for business intelligence and data visualization.
- Creates interactive reports and dashboards.
- Integrates with multiple data sources.
- Helps in decision-making through visual insights.

3. Google Data Studio

- Used to create customizable reports and dashboards.
- Connects with Google Analytics and other platforms.
- Provides real-time data visualization.
- Easy to share reports online.

5. Matplotlib

- Used for creating graphs and charts using Python.
- Supports line graphs, bar charts, scatter plots, etc.
- Useful for customized data visualization.

6. Seaborn

- Built on Matplotlib for advanced visualization.
- Provides attractive and statistical graphs.
- Useful for data analysis and presentation.

7. Excel

- Used for basic charts and data analysis.
- Supports bar charts, pie charts, and line graphs.
- Easy and widely used tool.

8. Sprout Social

- Provides built-in visualization dashboards.
- Tracks engagement and performance metrics.
- Helps analyze social media campaigns.

10. Google Analytics

- Tracks traffic from social media platforms.
- Provides visual reports and insights.
- Helps analyze user behavior.

CASE STUDY: SOCIAL MEDIA ANALYTICS ON FACEBOOK

7.1 Introduction

Social media platforms such as Facebook and X (formerly Twitter) play an important role in communication, marketing, and information sharing. These platforms generate a large volume of data every second through user activities like posting, liking, commenting, sharing, and following. This data can be analyzed using social media analytics to understand user behavior, engagement patterns, and content performance.

This case study focuses on analyzing social media data from Facebook and X to evaluate how organizations can use analytics to improve their marketing strategies and decision-making. The study highlights the importance of engagement metrics, trend analysis, and user interaction in improving social media performance.

7.2 Objectives

- To analyze user engagement on Facebook and X
- To study the performance of different types of content
- To identify trending topics and hashtags
- To understand user behavior and preferences
- To evaluate the effectiveness of social media strategies

7.3 Data Collection

Data collection is an important step in social media analytics. For this case study, data is collected from both Facebook and X platforms. On Facebook, data includes posts, reactions (likes, love, etc.), comments, shares, and page insights. On X, data includes tweets, reposts, likes, hashtags, and mentions. Data is collected using APIs, analytics dashboards, and publicly available data. The collected data includes both structured data (likes, shares) and unstructured data (comments, text posts).

7.4 Data Analysis

After collecting the data, analysis is performed to extract useful insights. Engagement rate is calculated using likes, comments, and shares. Trend analysis is performed using hashtags and popular topics. Sentiment analysis is used to understand user opinions as positive, negative, or neutral. Follower growth is analyzed over time to measure popularity. Content performance is compared based on type (text, image, video). This analysis helps organizations understand which content performs better and

how users interact with it. distribution channels, and promotional methods that would best reach their target audience.

7.5 Visualization

- ❖ Visualization techniques are used to represent the analyzed data in graphical form.
- ❖ Line graphs are used to show follower growth over time.
- ❖ Bar charts are used to compare engagement metrics across posts.
- ❖ Pie charts are used to represent audience distribution.
- ❖ Network graphs are used to identify influencers and user connections.
- ❖ Visualization makes it easier to understand patterns and trends in social media data.

7.6 Challenges

- There are several challenges faced during social media analysis:
- Large volume of data makes analysis complex.
- Presence of fake accounts and bots affects data accuracy.
- Unstructured data such as text and images is difficult to analyze.
- Privacy issues restrict access to certain data.
- Rapidly changing trends require continuous monitoring.

CONCLUSION

Social media analytics has become an essential tool in the modern digital world for understanding user behavior, analyzing trends, and supporting business decision-making. This report highlighted the importance of collecting, preprocessing, and analyzing social media data to generate meaningful insights. Through various techniques such as summary statistics, modeling, and data visualization, organizations can better understand audience engagement and content performance. The report also discussed different types of data in social media, analytical tools, and visualization methods that help in simplifying complex data. Ethical issues such as data privacy, user consent, and data misuse were also considered, which are important for responsible use of social media analytics.

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CONCLUSION

Social media analytics has become an essential tool in the modern digital world for understanding user behavior, analyzing trends, and supporting business decision-making. This report highlighted the importance of collecting, preprocessing, and analyzing social media data to generate meaningful insights. Through various techniques such as summary statistics, modeling, and data visualization, organizations can better understand audience engagement and content performance. The report also discussed different types of data in social media, analytical tools, and visualization methods that help in simplifying complex data. Ethical issues such as data privacy, user consent, and data misuse were also considered, which are important for responsible use of social media analytics.

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